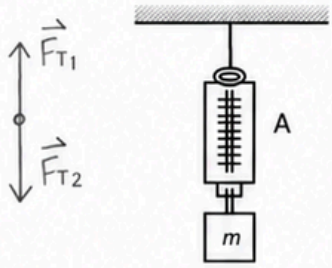
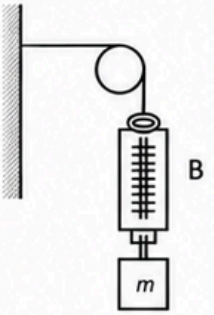
Learning Goals (Students will be able too…)

- Solve problems that have objects attached by a string hanging over a pulley .

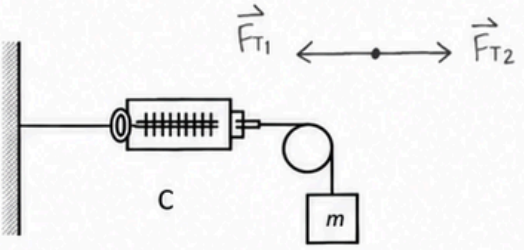
Tension & Pulleys

Suppose you were to hang a mass of **0.5 kg** in the various configurations shown below. You may assume the scales have no mass.





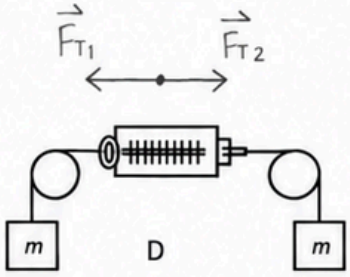


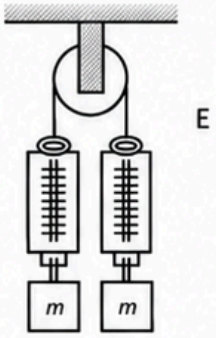


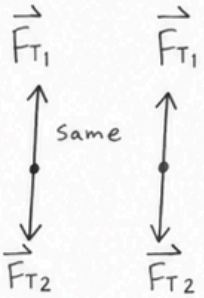
1. Draw a free body diagram for the spring scale itself in each scenario.

Every scale (A-E):

$F_{T1} = F_{T2} = 4.9$ N



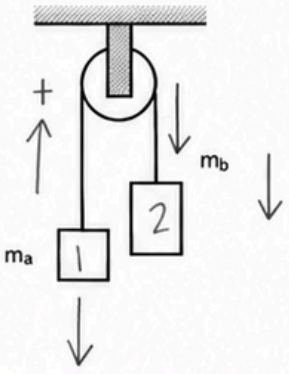




Atwood Machine

Typical pulley problems will consist of an Atwood Machine.

Atwood Machine – consists of two weights tied together and suspended over a pulley as the photo to the right depicts.



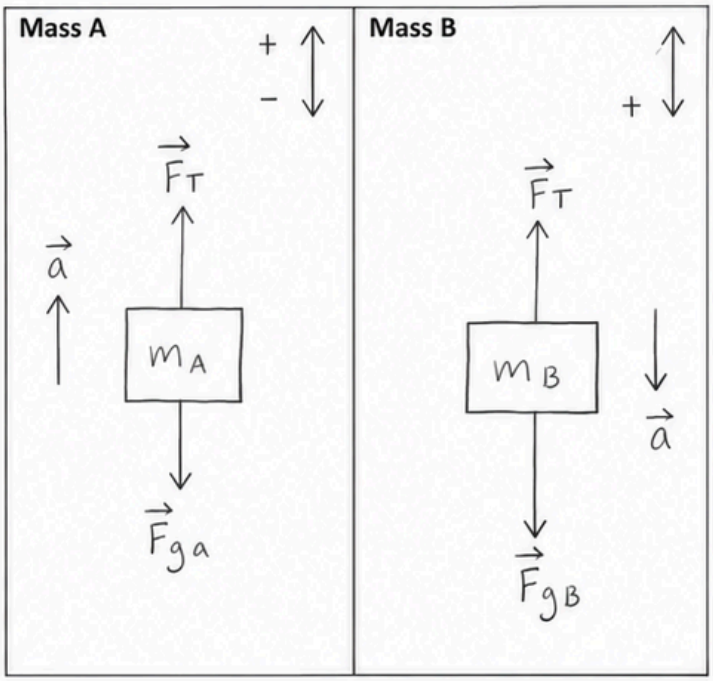
1) If the masses are equal, if we were to give one a gentle push to a mass and release it, what type of motion will occur **after** it is released?

-Uniform motion / constant velocity

2) Suppose that m_b is greater than m_a. Will mass A accelerate? If so, in what direction? Will mass B accelerate? If so, in what direction? How will those accelerations compare in magnitude? Explain.

Yes. A accelerates upward and B downward, with equal magnitudes because the string does not stretch. With each positive axis along its own acceleration, both scalar accelerations are positive.

3) Draw a FBD for mass A and for mass B.($m_b > m_a$). Indicate the acceleration of each mass and **choose your sign conventions so they align with the acceleration of the mass!** (positive will be different for each mass)



4) For the FBDs, do you need to use a different symbol for the magnitude of the acceleration of mass A and mass B (i.e. a_a and a_b)? What about for the forces of tension and gravity acting on each mass? Explain.

For acceleration and tension, the same symbol can be used because all the magnitudes are the same, but for gravity you need a different symbol because the masses are different.

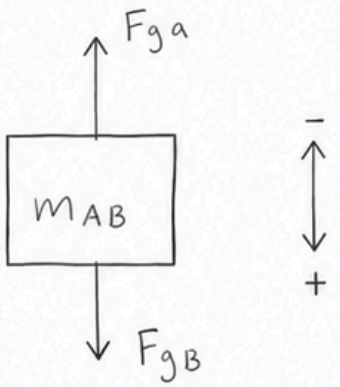
5) Now we must realize that both masses accelerate **together, as a system**.

a. What force(s) act to move the system forward?

Gravity on B ($\vec{F}_{gB}$)

b. What force(s) act to oppose the motion of the system?

Gravity on A ($\vec{F}_{gA}$)



Straightened system: F_{gA} opposes motion. (Actual gravity on A points down.)

c. Write an scalar equation for the net force on the system:

$F_{net} = F_{gB} - F_{gA}$

d. If $m_a = 0.200$ kg and $m_b = 0.250$ kg, determine the acceleration of the system:

$F_{net} = F_{gB} - F_{gA}$

$F_{net} = m_b g - m_a g$

$F_{net} = (0.25)(9.81) - (0.2)(9.81)$

$F_{net} = 0.4905\ N$

$a = \frac{F_{net}}{m_{AB}}$

$a = \frac{0.4905}{0.450}$

$a = 1.09\ m/s^2$

$a = +1.09\ m/s^2$

Calculate Acceleration - Alternative Method

1. Go back to your **FBD** of **mass A**. Use **Newton's 2nd Law** to write down an equation that relates the forces acting on **mass A** to the **acceleration** of **mass A**. Write the forces as scalars and use + or -- to indicate direction. (Don't solve for *a*.)

$$F_{net} = m_A a$$

$$F_{net} = F_T - F_{gA}$$

$$m_A a = F_T - F_{gA}$$

$$a = \frac{F_T - F_{gA}}{m_A}$$

2. Use **Newton's 2nd Law** to write down a **scalar equation** that relates the forces acting on **mass B** to the acceleration of **mass B**.

$$F_{net} = m_B a$$

$$F_{net} = F_{gB} - F_T$$

$$m_B a = F_{gB} - F_T$$

$$a = \frac{F_{gB} - F_T}{m_B}$$

3. Algebraically eliminate **T** from the above two equations and solve for the acceleration of the masses.

① $a = \dfrac{F_T - F_{gA}}{m_A}$

$m_A \times a = F_T - F_{gA}$

② $a = \dfrac{F_{gB} - F_T}{m_B}$

$m_B \times a = F_{gB} - F_T$

③ Add the equations

$(m_A \times a) + (m_B \times a) = (\cancel{F_T} - F_{gA}) + (F_{gB} - \cancel{F_T})$

$(m_A + m_B)(a) = F_{gB} - F_{gA}$

$a = \dfrac{F_{gB} - F_{gA}}{m_A + m_B} \rightarrow a = \dfrac{(9.81)(0.25) - (9.81)(0.2)}{0.25 + 0.2}$

$a = 1.09\ m/s^2$

4. Use an equation from question **8. or 9.** to solve for the magnitude of the force of tension. How does this value compare with F_{ga} and F_{gb}? Explain why.

$$a = \frac{F_{gB} - F_T}{m_B}$$

$$m_B \times a = F_{gB} - F_T$$

$$(m_B \times a) - F_{gB} = -F_T$$

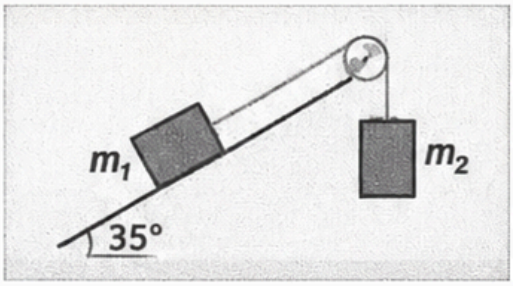
$$-(m_B \times a) + F_{gB} = F_T$$

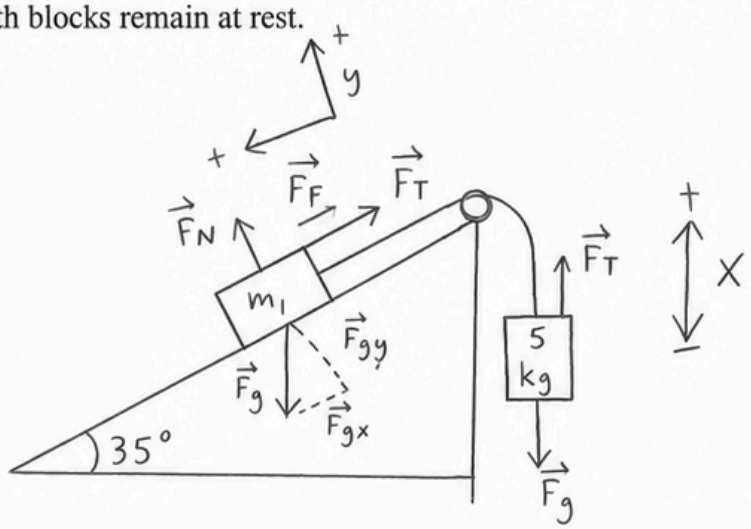
$$-(0.250 \times 1.09) + (0.250)(9.81) = F_T$$

$F_T = 2.18\ N$

Classroom Example

Example 1) The figure below shows two blocks joined with a rope that runs over a pulley. The mass of m_2 is $5.0kg$, and the incline is $35°$. The coefficient of static friction between m_1 and the inclined plane is 0.25. Determine the largest mass for m_1 such that both blocks remain at rest.





$F_{netx} = F_{g1x} - F_F - F_T$

$f_{netx} = 0$

$0 = F_{g1x} - F_F - m_2g$

$0 = F_{g1}\sin 35° - F_F - m_2g$

$0 = F_{g1}\sin 35° - (\mu)(F_N) - m_2g$

$0 = F_{g1}\sin 35° - (\mu)(F_{g1}\cos 35°) - m_2g$

$0 = m_1g\sin 35° - (\mu)(m_1g\cos 35°) - m_2g$

$0 = 9.81m_1\sin 35° - (0.25)(9.81m_1\cos 35°) - 49.05$

$49.05 = 9.81m_1\sin 35° - (0.25)(9.81m_1\cos 35°)$

$49.05 = 5.627m_1 - 2.009m_1$

$49.05 = 3.618m_1 \quad m_1 = 13.56\ kg$

$m_1 = 14kg$

For m_2

$F_{net} = 0$

$F_{net} = F_T - F_g$

$F_T = F_g$

$F_T = m_2g$

Find F_F using m_1

$F_{net_y} = 0$

$F_{net_y} = F_N - F_{g1y}$

$0 = F_N - F_{g1y}$

$F_N = F_{g1y}$

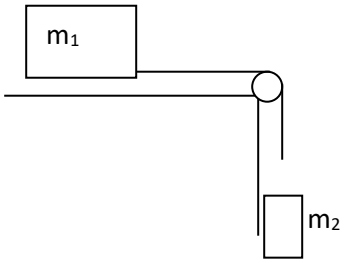
$F_N = F_{g1}\cos 35°$

Homework

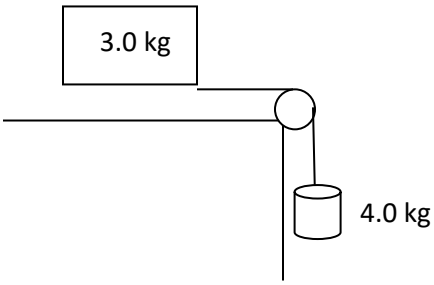
Txt pg 82 #6, pg 90 #6,8

Try these first :

1. In the diagram below, m_1 = 1.2 kg and m_2 = 0.500 kg. The two masses are connected by an inextensible string and m_1 slides along a frictionless surface. Find the acceleration of the system and the string tension. (2.9 m/s^2 3.5 N)



2. Find the tension in the string connecting the two blocks. Do not find the acceleration. There is 30 N of friction between the block and the table. The string is light and the pulley is frictionless. [34 N]



3. What is the value of *m* that would give the system an acceleration of 2.0 m/s/s? The coefficient of friction is 0.15, the string is light and the pulley is frictionless. [9.5 kg]

